Supplementary Material: Masked Synchronized Locator Evolution for Re-Keyable Private Face Identification

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I. Evidence and Version Boundaries

The original submission and first revision batch are frozen. The corrected materialized reference is `revision_v2/mpc/msle_system.mpc`, with independent oracle `revision_v2/reference.py`. The new virtual, AES and private-client variants have separate sources in `revision_v3/mpc`. Every run records the SHA-256 of the circuit, parameters, backend configuration and input artifact. Only measurements matching the indicated final source hashes enter each performance comparison. Warmups, diagnostic runs, earlier source versions and legacy circuits remain in the logs but are not pooled with final timed runs.

An audit found that the legacy integrated program sampled five positive 30-bit key contributions and added them. The sum has at most $\log_2(5(2^{30}-1)+1) \simeq 32.32$ bits of support, not 128 bits. The legacy circuit also used 73 MiMC-style rounds and an implicit-field/bounded-output-decomposition configuration. Its timings cannot establish the cost of the corrected cryptographic configuration. We replace them with a known-prime implementation, independent uniform full-field contributions, 81-round permutations and canonical 128-bit output decomposition. The old program is preserved for audit, not silently relabeled as secure evidence.

The unmasked shared-root baseline is an internal, unpublished implementation. Frozen source `sle_sort_merge_transition.mpc`, the integrated circuit and their configuration files preserve its provenance. The revised sort/merge comparator is adapted to the same complete masked-state contract, field width, capacity and verification as the direct transition. Reported comparison timings use this matched configuration.

II. Concrete Cryptography and Independent Tests

`parameters.json` contains the decimal prime, all 81 constants, generation domain and assumptions for the MiMC construction [1]. The script `init_parameters.py` verifies primality and $\gcd(3, p-1) = 1$. Keys are sampled with operating-system randomness using rejection-based `secrets.randbelow(p)` for each party and each of nine keys. Summing modulo p is uniform if at least one contribution is independent and uniform. The passive model does not require malicious input-validation machinery.

The locator key is independent of eight mask-word keys. For each mask word, apply its MiMC permutation to the locator, decompose the field element canonically into 128 bits, and retain bits 0–63. All eight words are concatenated in the released bit order. No additive offset is used as an unproven domain separator between keyed calls. The independent oracle implements both encryption and inversion and generates full-field vectors including 0, $p-1$, 2^{64} and 2^{127}.

The secure primitive test checks 16 values and 1024 selected mask bits, exact full-field equality, row-association preservation under secret shuffle and the shuffled multiset. Separate reference unit tests verify permutation inversion, key-support configuration and exact re-keying scores. Passing these tests establishes implementation agreement, not a cryptanalytic proof of MiMC.

The SHAKE-256 lifecycle simulator is separate: its domain is `M-SLE/root-mask/v1`, followed by the key bytes and an eight-byte big-endian simulation locator. These historical simulation locators are 64 bits. Corrected MPC locators are uniform field elements near 128 bits. Attack diagnostics from the simulator are labeled as such and never used to infer MPC performance or exposed-key security.

III. Persistent State and Access Schedule

For N records, $d = 512$, block width $w = 4$, and capacity $C = \min(N, 8 + \lceil N/4 \rceil)$, there are $M = (d/w)2^w C$ padded slots. The logical state per party contains

$$S = N(d+3) + 2M + 9$$

field shares: locator, label, tie tag and d protected bits per database row; locator and validity per slot; nine keys. Its nominal 128-bit payload is $16S$ bytes per party. MP-SPDZ serialization, object overhead and communication buffers are additional. The shared-root baseline has one key and $S - 8$ fields.

Database rows are shuffled as complete bundles of $d + 3$ fields. Labels and tie tags therefore do not remain in public row order. The index retains a public physical bucket/slot schedule, but locator and validity contents remain secret-shared.

Every lookup scans every slot and uses secret flags. No queried bucket, number of candidates or fallback bit is opened. The deliberate price is scanning and comparing the entire padded index. This is a concrete access-hiding reference backend, not an implementation of Path ORAM or a sublinear search claim.

After a transition, every field is refreshed by adding independently input zero sharings from all five parties. The represented value remains unchanged. We rely on the passive Shamir input/refresh protocol; functional refresh tests do not establish a composable mobile-security guarantee. Secure erasure and the corruption-period definition remain external assumptions.

The persisted-state family reads each server's own share file at fixed public offsets. An external trusted initialization fixture samples two full-field coefficients per state field and sends $f(j)$ only to server j, $j = 1, \ldots, 5$. MPC privately inputs these values and interpolates with weights $(5, -10, 10, -5, 1)$. The client checks this reconstruction for every field before installation. Thus a computing party is given one Shamir evaluation, rather than the plaintext database or bucket contents. The fixture knows the public research roots and initial keys and is outside the corruption model. It is a test initializer, not a deployed capture application. Each later re-key receives only new key contributions and refresh inputs; queries receive a probe and threshold. Old keys, templates, locators, validity bits, labels and tie tags come from the persisted share bundle.

The input-inclusive reference initializes invocations independently; the persisted-state family instead performs successive transitions and reload queries on the same logical store. The two families are not pooled. A source audit verifies identical cryptographic helpers and protocol tails; their state entry points differ. Tests cover exact/reject decisions, private indexed retrieval, and an omitted candidate entry followed by rollback and a separate reload query. MP-SPDZ signals a successful read ending exactly at EOF with -1; short reads raise an I/O error. The adapter distinguishes these conditions.

The write instruction stores one logical selected-state bundle. Filesystem crash recovery, distributed publication and operating-system secure erasure are outside the implementation; local audit files are not a mobile-secure production store. A rejected relation retains old decoded values and keys even if maintenance changes shares or physical row order. Accordingly, rollback tests assert semantic state equality, not byte equality of files.

IV. Retained and Reproduced Lifecycle Diagnostics

The released 128-identity CelebA simulation uses 20 epochs and seed 20260818. Each mask key is 32 bytes. Initial locators are sampled without replacement from $1, \ldots, 2^{31} - 1$ and serialized in eight bytes. Subsequent simulation locators are 64-bit keyed BLAKE2b tokens with an explicit collision check. This hash-based token surrogate is not a proof or implementation of a permutation on all 64-bit inputs. Corrected MPC uses the actual full-field permutation described above.

At every adjacent-epoch transition, direct-linkage AUC compares 128 mated negative-Hamming scores with $128 \cdot 127$ non-mated scores. Hungarian assignment minimizes the complete old/new Hamming cost after randomizing observed new-row order. The relational attack excludes the self-distance, sorts the remaining 127 within-epoch distances for each row, forms the mean absolute difference between every old/new signature pair, and applies the same assignment algorithm. A single global epoch XOR mask is the relational positive control. Stable templates give the direct-linkage positive control; supplying the correct key and locator recovers the root exactly. None of these controls uses the corrected MPC timings.

A fixed random 75/25 identity split gives 96 training and 32 test rows for a multi-output ridge attack. Inputs are protected bits with a bias coordinate, outputs are root bits, ridge parameter is 10, and predictions are thresholded at 0.5. The comparison predicts every bit from its training-set majority. The observed attack/prior accuracies, 0.5242/0.5627, concern this attack family and auxiliary-information setting only. Per-epoch means and the original descriptive normal intervals are preserved; they are not uncertainty over new biometric populations.

The separate refresh demonstration uses toy field 65537 on 65536 bit positions. Three same-epoch shares reconstruct, while three shares drawn from different refreshed sharings coincide at one position. This is a functional illustration, not a 128-bit cryptographic benchmark or a mobile-adversary proof. The actual corrected MPC refresh uses the main field and the entire state bundle.

The revision reruns the released simulation with one resource-only correction: integer-population sampling replaces a materialized 16-GiB integer range. The sampled indices and RNG state are unchanged. Original and corrected source, expected aggregates and the reproduced per-epoch outputs remain in the artifact. This avoids making ordinary reproduction depend on a machine with more than 16 GiB available RAM.

V. Retrieval, Capacity and Integrity

For each block g, form secret one-hot flags for the query's w-bit value. Multiply each slot validity by its bucket flag. For every database row, compare its locator with every slot using full-field equality; OR the flagged matches to deduplicate. Calculate exhaustive and candidate-restricted winners with the same secret stable tie tags. If not all records were retrieved and the candidate winner's distance is at least d/w, privately select the exhaustive winner. All scores and both selection paths are computed on a public schedule.

Enrollment counts are checked before executing a benchmark. If any bucket exceeds C, the run rejects enrollment and requires a new public capacity configuration. CPU diagnostic tables also show unbounded candidate unions and deliberately

insufficient capacities, explicitly flagged. Such rows are not results from silently truncated secure searches. The eight-bit capacity policy is $\min(N, 8 + \lceil 4N/2^8 \rceil)$ and often fails at larger N; its unbounded candidate recall is diagnostic only.

Expected relation tuples are derived from the *proposed* decoded templates. In addition, every proposed template is compared with the old protected template XOR the old and new masks. The relation acceptance bit is the product of template binding, locator-translation success and two independent multiset checks. This correction binds both old/new roots and new root/index membership. Invalid slots contribute the multiplicative identity. Two independently sampled full-field challenge triples follow candidate fixation.

Fault cases are: one missing valid occurrence; duplicate occurrence; move to the wrong bucket; corrupt a locator; use a wrong index-transition key; retain one old locator; flip one protected-template bit; retain one old protected template. For each case the diagnostic circuit verifies rejection, old-state retention and a subsequent valid query. Accept/reject controls use threshold zero with an exact enrolled root and a distinct probe. These checks are application integrity tests within passive MPC.

VI. Biometric Selection and Calibration

The submitted CelebA NPZ remains byte-identical (SHA-256 `96862a03...486f0db1`). Source images are recovered from the public `flwrlabs/celeba` dataset pinned to revision `2d738f56e0e7f925ea36ae7c808ea925264aacec`. A range reader downloads only the identity columns of 19 Parquet shards; row indices identify the first six train images for the same 128 evaluation identities. Image bytes and shapes are checksummed. Recovered floats are compared with the fixed gallery/probe root bits and with all historical split metrics. Any mismatch is reported explicitly and the new float/binary pair remains valid only for its reconstructed image protocol.

The main fixed CelebA cohort and the historical portability cohort are distinct. The latter samples 128 of the available train identities with at least six images using seed 20260722; only one identity overlaps the fixed cohort. We therefore reconstruct its original images independently before comparing the 20 historical ranks. A direct comparison of new splits on the fixed cohort with the old portability scores is retained as an audit of this population distinction, not interpreted as failed algebraic reproduction. The first six original Parquet image rows per identity are used; unavailable thumbnail URLs are not substituted with resized images.

LFW extraction uses the complete funneled archive, not an incomplete extraction directory or stale joblib cache. There are 13233 JPEGs and 5749 identities; 311 have at least six images. The old 192-identity manifest supplies 128 evaluation, 32 pilot calibration and 32 pilot unknown identities. The 20 split seeds are `20260722+20000+split`; two of six images form a gallery mean and the remaining four are probes. The reconstructed binary Rank-1 and Rank-5 equal every historical LFW split exactly.

The additional manifest freezes seed 20261030 before extracting 5557 new identities. The first 512 permuted identities calibrate; the next 512 are unknown tests; the rest are distractors. A random image is chosen using this same frozen generator, without quality or recognition filtering. All new cohorts exclude the original 192 identities. The same crop and model preprocessing are applied to every cohort: LFW crop $(78, 70, 172, 195)$, 94×125 intermediate image, 160×160 FaceNet input, RGB normalization $(x - 0.5)/0.5$, VGGFace2 pretrained InceptionResnetV1, normalized output embeddings. Complete image IDs and versions are in the manifests. These choices reproduce the historical pipeline; no claim is made that this is an optimal modern face frontend.

At each gallery size, unknown scores are the maximum cosine similarity or negative Hamming distance over the same gallery used for enrolled tests. Sort 512 calibration maxima in ascending order. With allowed count k, choose the next representable number strictly above the $(k+1)$th highest score. Accept when score is at least threshold. This handles tied binary scores conservatively. The empirical rule uses $k = 5$; the conservative rule uses the largest k with

$$\Pr\{\mathrm{Binom}(512, 0.01) \le k\} \le 0.05,$$

which is $k = 1$. This is an order-statistic statement under IID identity-score sampling, not a bound under population shift. The unknown test cohort is never used for threshold selection.

Verification-pair pilot rows additionally compare four probes per enrolled identity with every gallery prototype, giving one genuine and $N - 1$ impostor pairs per probe. Their thresholds use a separate 32-identity calibration gallery. Pair counts do not imply pair independence. The main revised rejection evidence instead uses independent unknown identities and identity-level intervals. The earlier frequent-identity pilot (at least six source images per identity, six probes each) is retained as a harder, distinct population. It must not be pooled with the new single-image unknown cohort to imply a stable low FPIR.

VII. Secure Measurement and Cost Model

All five party processes share one WSL host, over loopback. The frozen MP-SPDZ [2] source revision begins `9d809599`; its full 40-character commit is recorded in the artifact. Ubuntu uses g++ 13.3.0. The host is Windows 11 Home Single Language 10.0.26200, Intel i5-1135G7, four cores/eight threads, 21154324480 bytes RAM; WSL is limited to about 9.6 GiB RAM and 3 GiB swap. Build and runtime logs retain exact commands. The compiler receives `-P p -b 64 -M`; the circuit sets statistical security 80, bit length 64 for bounded scores/group identifiers, and explicitly 128 for field output decomposition and sorting.

`-b 64` is the compiler optimization budget, not the field size. Locator permutation, relation-factor and state-selection vectors are batched at 4096; canonical mask decomposition uses batches of 64 records. The compiler memory budget is 512, and radix sorting uses one worker thread. Public runtime loops bound compiler expansion without changing key separation or field arithmetic. Earlier unrolled-mask and unbounded-broadcast versions hit compiler limits at $N = 128$ and $N = 256$ respectively; their outputs are preserved and excluded from final performance results.

Each performance point in the retained materialized-reference batch has a warmup followed by three recorded repetitions with fresh keys and varying probe rows. New persisted, cipher and private-client comparisons use five measured repetitions; larger cold-input and network-delay comparisons use three. Their ordering, warmups and source hashes are retained per series. Oracle inputs and diagnostic comparisons are disabled in timed circuits; each query result is checked by the external controller, with private-client results decoded only there. Compile time is logged separately. Total VM time includes input and maintenance, and wall time also includes launcher overhead. Each timer's communication is party-zero traffic; *global* traffic is a separate total from MP-SPDZ and is never mislabeled as a phase total. RSS from `/usr/bin/time -v` is the largest child's peak, not aggregate five-party memory.

The optimized sort/merge baseline first stably compacts the M padded rows by their secret validity bit using a one-bit oblivious radix sort. The known valid count is $L = N(d/w)$; a private check verifies that the first L rows are valid and the remaining rows are invalid. Only the $N + L$ database and valid-index rows are then sorted by all 128 canonical old-locator bits. Stable order puts database source rows before index rows within an equal-key group. Full-field adjacent equality produces bounded group IDs; segmented doubling propagates the new source locator. Perfect-permutation reveal-sort operations restore join positions and then all original padded slots; they do not open the hidden locator ordering. Invalid slots receive fresh field values. This avoids both sorting every dummy slot at full field width and overflowing the field with a packed 129-bit key. All template masking, verification, logical selection, shuffle and refresh match the direct transition, as verified by a source audit. Insufficient capacity aborts both equally. The interrupted all-padding-sort warmup has no completed timing result and is excluded.

The held-out cost model and communication projections use only the input-inclusive reference family. They do not predict persisted-file loading cost. The two families have the same computation after loading, but their initialization and I/O paths differ; their latency observations are therefore reported separately.

A. Complete Repetition and Phase Tables

TABLE I
PERSISTED-STATE PHASE MEANS. READ/WRITE EXECUTES ON ALL FIVE LOCAL SHARE FILES; TRAFFIC IS PARTY-ZERO MB. MODE1 RE-KEYS, MODE2 QUERIES.

N	State	Mode	Phase	Seconds	P0 MB
8	Masked	1	Read own shares	0.009	0.000
8	Masked	1	Locator/index	5.219	84.976
8	Masked	1	Old/new masks	3.531	3.120
8	Masked	1	Verification	0.816	2.678
8	Masked	1	Selection	0.082	1.180
8	Masked	1	Shuffle/refresh/write	1.049	2.022
8	Raw	1	Read own shares	0.009	0.000
8	Raw	1	Locator/index	5.880	84.976
8	Raw	1	Old/new masks	0.000	0.000
8	Raw	1	Verification	0.750	2.547
8	Raw	1	Selection	0.060	1.180
8	Raw	1	Shuffle/refresh/write	1.120	2.022
8	Masked	2	Read own shares	0.011	0.000
8	Masked	2	Fresh query masks	2.040	1.852
8	Masked	2	Hamming/ArgMin	0.183	0.280
8	Raw	2	Read own shares	0.009	0.000
8	Raw	2	Hamming/ArgMin	0.385	0.182
128	Masked	1	Read own shares	0.073	0.000
128	Masked	1	Locator/index	24.408	425.337
128	Masked	1	Old/new masks	7.168	43.652
128	Masked	1	Verification	1.686	21.094
128	Masked	1	Selection	0.347	7.344
128	Masked	1	Shuffle/refresh/write	7.029	45.397
128	Raw	1	Read own shares	0.089	0.000

TABLE II
PERSISTED-STATE PHASE MEANS. READ/WRITE EXECUTES ON ALL FIVE LOCAL SHARE FILES; TRAFFIC IS PARTY-ZERO MB. MODE1 RE-KEYS, MODE2 QUERIES.

N	State	Mode	Phase	Seconds	P0 MB
128	Raw	1	Locator/index	21.852	425.337
128	Raw	1	Old/new masks	0.001	0.000
128	Raw	1	Verification	1.227	18.996
128	Raw	1	Selection	0.309	7.344
128	Raw	1	Shuffle/refresh/write	6.839	45.397
128	Masked	2	Read own shares	0.061	0.000
128	Masked	2	Fresh query masks	5.854	20.045
128	Masked	2	Hamming/ArgMin	0.701	4.509
128	Raw	2	Read own shares	0.117	0.000
128	Raw	2	Hamming/ArgMin	1.083	2.976

TABLE III
PERSISTED-STATE MEMORY AND LAUNCHER-INCLUSIVE TIMING; LARGEST-CHILD RSS IS NOT AGGREGATE MEMORY.

N	State	Mode	Wall s	Peak RSS MiB	State MiB
8	Masked	1	11.815	39.58	0.563
8	Raw	1	8.853	33.65	0.563
8	Masked	2	2.974	28.61	0.563
8	Raw	2	1.179	22.43	0.563
128	Masked	1	42.736	137.49	3.506
128	Raw	1	31.660	102.80	3.506
128	Masked	2	7.975	68.00	3.506
128	Raw	2	2.280	41.52	3.506

TABLE IV
VALID-COMPACTED FULL-WIDTH SORT/MERGE: REPETITION RANGE, WALL TIME AND MEMORY. THREE MEASURED RUNS AFTER A WARMUP; INPUT-INCLUSIVE FAMILY.

N	Wall s	VM range s	Rounds	Peak RSS MiB	State MiB
8	94.86	74.82–118.72	54793	151.62	0.563
16	128.35	123.99–132.54	66300	193.19	0.876

TABLE V
VALID-COMPACTED SORT/MERGE PHASE MEANS; TRAFFIC IS PARTY-ZERO MB.

N	Phase	Seconds	P0 MB
8	Compact/sort/translate/restore	90.782	1541.040
8	Old/new masks	1.659	2.782
8	Verification	0.404	2.571
8	Selection	0.042	1.180
8	Shuffle/refresh/write	0.804	2.022
16	Compact/sort/translate/restore	124.512	3345.350
16	Old/new masks	1.193	4.584
16	Verification	0.347	4.603
16	Selection	0.056	1.836
16	Shuffle/refresh/write	0.971	4.261

TABLE VI

REPETITION RANGE, LAUNCHER-INCLUSIVE MEAN WALL TIME, PARTY-ZERO ROUNDS AND NOMINAL PERSISTENT STATE PER PARTY. RUNTIME TABLES USE MB AS PRINTED BY MP-SPDZ.

N	Design	Operation	Runs	Wall s	VM range s	Rounds	State MiB
8	Masked	Re-key	3	10.57	8.36–10.85	8181	0.563
8	Raw	Re-key	3	6.39	5.44–5.94	4855	0.563
8	Masked	Query	3	3.06	2.22–2.32	2038	0.563
8	Raw	Query	3	1.15	0.35–0.66	541	0.563
8	Masked	Indexed query	3	43.86	42.58–44.24	25080	0.563
16	Masked	Re-key	3	10.78	9.72–10.42	10453	0.876
16	Masked	Query	3	2.42	1.51–2.00	2150	0.876
32	Masked	Re-key	3	13.15	11.32–14.13	13117	1.252
32	Raw	Re-key	3	11.68	10.07–12.31	9771	1.251
32	Masked	Query	3	2.73	1.85–2.54	2330	1.252
32	Raw	Query	3	1.06	0.46–0.52	821	1.251
64	Masked	Re-key	3	17.56	16.55–16.89	18438	2.003
64	Masked	Query	3	3.06	2.24–2.38	2642	2.003
128	Masked	Re-key	3	29.58	25.81–30.09	32116	3.506
128	Raw	Re-key	3	23.19	22.05–23.09	25650	3.506
128	Masked	Query	3	4.80	3.81–4.49	4746	3.506
128	Raw	Query	3	1.51	0.87–0.96	1681	3.506
256	Masked	Re-key	3	45.21	43.68–44.30	59455	6.512
256	Masked	Query	3	9.12	8.01–8.82	8916	6.512

TABLE VII

PHASE MEANS; COMMUNICATION IS PARTY-ZERO TRAFFIC, NOT GLOBAL TRAFFIC. MODE 1 IS RE-KEY, 2 FULL-SCAN QUERY, 3 PRIVATE INDEXED QUERY.

N	Design	Mode	Phase	Seconds	P0 MB
8	Masked	1	Locators/index	4.697	84.976
8	Masked	1	Old/new masks	3.236	3.120
8	Masked	1	Verification	0.563	2.678
8	Masked	1	Selection	0.055	1.180
8	Masked	1	Shuffle/refresh/write	1.058	2.022
8	Raw	1	Locators/index	4.217	84.976
8	Raw	1	Verification	0.424	2.547
8	Raw	1	Selection	0.046	1.180
8	Raw	1	Shuffle/refresh/write	0.852	2.022
8	Masked	2	Fresh query masks	2.085	1.852
8	Masked	2	Hamming/ArgMin	0.162	0.280
8	Raw	2	Hamming/ArgMin	0.471	0.182
8	Masked	3	Fresh query masks	0.654	1.875
8	Masked	3	Hamming/ArgMin	0.077	0.299
8	Masked	3	Private candidate retrieval	42.396	990.576
128	Masked	1	Locators/index	16.222	425.337
128	Masked	1	Old/new masks	5.169	43.652
128	Masked	1	Verification	0.931	21.094
128	Masked	1	Selection	0.257	7.344
128	Masked	1	Shuffle/refresh/write	4.852	45.397
128	Raw	1	Locators/index	16.008	425.337
128	Raw	1	Old/new masks	0.001	0.000
128	Raw	1	Verification	0.764	18.996
128	Raw	1	Selection	0.217	7.344

TABLE VIII
PHASE MEANS; COMMUNICATION IS PARTY-ZERO TRAFFIC, NOT GLOBAL TRAFFIC. MODE 1 IS RE-KEY, 2 FULL-SCAN QUERY, 3 PRIVATE INDEXED QUERY.

N	Design	Mode	Phase	Seconds	P0 MB
128	Raw	1	Shuffle/refresh/write	4.342	45.397
128	Masked	2	Fresh query masks	3.385	20.045
128	Masked	2	Hamming/ArgMin	0.398	4.509
128	Raw	2	Hamming/ArgMin	0.551	2.976
256	Masked	1	Locators/index	25.508	765.739
256	Masked	1	Old/new masks	6.904	87.004
256	Masked	1	Verification	1.338	41.017
256	Masked	1	Selection	0.388	13.640
256	Masked	1	Shuffle/refresh/write	7.889	102.360
256	Masked	2	Fresh query masks	7.100	40.092
256	Masked	2	Hamming/ArgMin	0.562	9.019

TABLE IX
INPUT-INCLUSIVE HELD-OUT COST-MODEL VALIDATION. TRAINING SIZES ARE ALWAYS 8,16,32. NO HELD-OUT RESULT IS REFITTED; PERSISTED-FILE LOADING IS OUTSIDE THIS MODEL.

Mode	Metric	Held-out N	Predicted	Measured	Error (%)
1	VM s	64	15.020	16.768	10.43
1	VM s	128	20.801	28.539	27.11
1	VM s	256	32.364	44.021	26.48
1	Traffic MB	64	1573.604	1564.010	0.61
1	Traffic MB	128	2718.928	2717.360	0.06
1	Traffic MB	256	5009.575	5051.490	0.83
2	VM s	64	1.847	2.297	19.62
2	VM s	128	1.597	4.087	60.93
2	VM s	256	1.097	8.346	86.86
2	Traffic MB	64	62.919	63.413	0.78
2	Traffic MB	128	122.368	122.499	0.11
2	Traffic MB	256	241.265	244.993	1.52

TABLE X
INPUT-INCLUSIVE COMMUNICATION PROJECTIONS ONLY, NOT EXECUTED MPC MEASUREMENTS. RUNTIME IS NOT EXTRAPOLATED. PADDING, FIELD AND BATCH POLICY ARE HELD FIXED.

Mode	Gallery N	Capacity	Index slots	Projected MB
1	1024	264	540672	18753.5
1	4096	1032	2113536	73729.0
2	1024	264	540672	954.7
2	4096	1032	2113536	3808.2

These projections use only the communication model that passed held-out validation below 5% relative error. Their out-of-range uncertainty remains unmeasured. They do not predict indexed-query cost, latency, throughput or a different ORAM backend.

B. CelebA Recovery Outcome

The recovered gallery has 0 binary bit differences and the recovered probes have 0 relative to the frozen NPZ. These counts determine whether the historical fixed split is exactly reproduced; the paired float/binary experiment always uses identical reconstructed images. Source and feature hashes remain in the artifact.

The distinct historical portability cohort reproduces both Rank-1 and Rank-5 exactly in 20 of 20 splits. All per-split comparisons are recorded; any mismatch is retained. The two CelebA populations are not pooled.

TABLE XI
SEPARATE CELEBA 20-SPLIT PAIRED COHORTS; INTERVALS RESAMPLE 128 IDENTITIES. DELTA IS BINARY MINUS FLOAT RANK-1 IN PERCENTAGE POINTS.

Cohort	Float R1	Binary R1	Delta [95% CI]
Fixed cohort, new splits	0.8822	0.8605	-2.168 [-2.842, -1.523]
Historical portability	0.8762	0.8515	-2.471 [-3.281, -1.729]

TABLE XII
OBSERVED TEST FPIR AND DIR WITH IDENTITY-BOOTSTRAP 95% INTERVALS, IN PERCENTAGES.

Gallery	Root	Rule	FPIR [95% CI]	DIR [95% CI]
128	Float	Emp.	1.602 [0.869, 2.461]	89.26 [86.42, 91.87]
128	Float	Cons.	0.371 [0.068, 0.762]	79.92 [75.93, 83.73]
128	Binary	Emp.	1.885 [1.152, 2.715]	87.76 [84.67, 90.66]
128	Binary	Cons.	0.537 [0.225, 0.957]	80.01 [76.20, 83.68]
4096	Float	Emp.	1.953 [0.781, 3.320]	71.70 [67.33, 76.03]
4096	Float	Cons.	0.420 [0.000, 1.006]	53.83 [48.95, 58.81]
4096	Binary	Emp.	1.172 [0.391, 2.148]	59.12 [54.39, 63.73]
4096	Binary	Cons.	0.391 [0.000, 0.947]	46.33 [41.58, 51.24]

The fresh unknown cohort contains 365 identities with one source image, 134 with two to five, and 13 with at least six. Image-count-stratified FPIR is exported for every split in the CSV. The strata are descriptive and were not used to tune thresholds.

TABLE XIII
DESCRIPTIVE TEST FPIR AT GALLERY 128 BY SOURCE IDENTITY IMAGE COUNT, IN PERCENTAGES. THE STRATA HAVE 365, 134 AND 13 IDENTITIES; THE SMALL FINAL STRATUM DOES NOT SUPPORT A LOW-FPIR GUARANTEE.

Root	Rule	One	Two–five	Six or more
Float	Emp.	1.644	1.642	0.000
Float	Cons.	0.205	0.858	0.000
Binary	Emp.	1.575	2.799	1.154
Binary	Cons.	0.370	1.045	0.000

VIII. RETAINED REFERENCE REPRODUCTION COMMANDS

From `revision_v2` in the configured environment, run:

```
python3 -m unittest test_reference.py
python3 mpc_driver.py --phase primitives
```

The first command checks clear parameter/algebra properties; the second tests full-field secure primitives. The script `run_secure_batch.py` executes sequentially to avoid concurrent overwrites of Player-Data. The script `evaluate_lfw_extended.py` regenerates all paired, open-set and retrieval CSVs from frozen features and manifests. The script `recover_celeba.py` has separate label, image-download and extraction phases; `recover_celeba_external.py` recovers the distinct historical 20-split cohort. The script `build_revision_results.py` regenerates retained-reference tables, filtering final hashes. For the current revision, `README_FINAL.txt` distinguishes data reconstruction, frozen-model evaluation, sequential MPC runs and summary-only table reproduction. Local source and raw logs accompany the revision; no remote repository publication is implied.

IX. NEW IMPLEMENTATION AND EVIDENCE AUDIT

The new sources and manifests are in revision_v3. The reference and all development attempts remain available. Warmups and diagnostic oracles are excluded from production timings. Cold protected-owner inputs, actual per-party state shares and external private query shares are distinct interfaces.

A numerical audit found that NumPy2 weak scalar promotion rounded a strict nextafter64 threshold when compared with float32 Hamming scores. Corrected evaluators convert the exact integer scores to float64 before thresholding. Frozen models, seeds, cohorts, calibration rules and all closed-set flags are unchanged. Every corrected native float/sign open-set flag exactly reproduces the original independent evaluator, and every calibration acceptance count respects its prescribed limit. Original attempts are retained; only corrected open-set outputs are used below.

X. VIRTUAL-STATE AND PRIVATE-CLIENT INTERFACES

Distinct secret handles preserve a complete root/index relation under an injective semantic-locator map. The accepted virtual transition changes keys and protected templates, then shuffles complete rows and refreshes every share, while logical handles remain stable. Thus it is not physical replacement of every logical reference. Hiding handles, index contents, labels and tie tags is essential. Proposed templates are checked against both the old/new mask equation and the proposed decoded root/index relation; challenges follow candidate fixation.

For client input, each bit, threshold and fresh output pad uses independently sampled degree-two coefficients. The five evaluations are privately input and interpolated inside MPC using weights $(5, -10, 10, -5, 1)$. Any two original evaluations

TABLE XIV
FACENET FULL PREDEFINED PROJECTION STUDY AT 4096: ALL WIDTHS/SEEDS/CENTERING VARIANTS. PUBLIC HYPERPLANES ARE ESTABLISHED PREPROCESSING.

Method	Rank-1 (%)	Rank-5 (%)
original_cosine	95.312	98.203
original_sign_512	93.672	97.607
centered_cosine	95.400	98.281
centered_sign_512	93.604	97.666
RP_original_512_seed20261030	92.402	97.256
RP_original_512_seed20261031	92.002	97.002
RP_original_1024_seed20261030	94.062	98.076
RP_original_1024_seed20261031	94.082	97.676
RP_original_2048_seed20261030	94.717	98.066
RP_original_2048_seed20261031	94.805	98.115
RP_centered_512_seed20261030	92.305	97.490
RP_centered_512_seed20261031	92.480	96.650
RP_centered_1024_seed20261030	94.062	98.066
RP_centered_1024_seed20261031	94.170	97.373
RP_centered_2048_seed20261030	94.453	97.988
RP_centered_2048_seed20261031	94.922	98.076

TABLE XV
SFACE FULL PREDEFINED PROJECTION STUDY AT 4096: ALL WIDTHS/SEEDS/CENTERING VARIANTS. PUBLIC HYPERPLANES ARE ESTABLISHED PREPROCESSING.

Method	Rank-1 (%)	Rank-5 (%)
original_cosine	98.604	98.887
original_sign_128	91.045	95.830
centered_cosine	98.535	98.848
centered_sign_128	90.469	95.801
RP_original_512_seed20261030	97.188	98.271
RP_original_512_seed20261031	96.748	98.096
RP_original_1024_seed20261030	98.018	98.701
RP_original_1024_seed20261031	97.744	98.623
RP_original_2048_seed20261030	98.330	98.779
RP_original_2048_seed20261031	98.086	98.730
RP_centered_512_seed20261030	97.295	98.389
RP_centered_512_seed20261031	96.895	98.252
RP_centered_1024_seed20261030	97.900	98.652
RP_centered_1024_seed20261031	97.861	98.691
RP_centered_2048_seed20261030	98.271	98.799
RP_centered_2048_seed20261031	98.223	98.760

are independent of the secret constant. Only $M = y + R \pmod p$ is opened, for identity/reject encoding y and uniform fresh pad R. For any fixed y, M is uniform; the client knows R and recovers y. There is no separate acceptance opening. This is a standard pad/MPC composition with honest binary input and private-channel assumptions. The benchmark controller acts as the external trusted client; it knows the probes/pads/reference answers and is not a computing-server corruption. Diagnostic oracle inputs are excluded from production privacy claims.

Exact polynomial lookup uses the Lagrange basis on the distinct valid handle support; invalid slots are suppressed by zero validity flags. Secret bucket selection still scans every padded slot. Its preconditions are a valid stored relation, distinct handles and bit-valued inputs, rather than malicious input validation. Candidate bits are checked against independent full-field equality. In a complete matcher the same flags equal the OR of exact block matches already available from Hamming mismatches; a full exact scan is the stronger minimal control. All metadata addresses remain public fixed scans.

XI. RETAINED REFERENCE TIMING CONTROLS

The following matched reference-batch results retain shared-root, materialized-mask and compact full-field sort controls. They are not pooled with the fresh virtual or private-client timing series. Their complete source/config/input hashes and raw logs identify the configuration. Root confidentiality below threshold already holds for shared roots; extra masking addresses only the stated separate value-exposure view.

XII. RESOURCES FOR THE NEW COMPLETE-SYSTEM COMPARISONS

Echo-calibration medians in milliseconds (target: observed): 0: 0.075; 1: 2.763; 20: 20.828; 50: 50.977. All 20 echo samples per target, relay CPU, queue high-water marks and per-party logs are retained. Total buffer backpressure events: 0; maximum

TABLE XVI
FROZEN-MODEL TRANSFER ON IDENTICAL IMAGES/SPLITS. CELEBA COHORTS SHARE ONE IDENTITY; DO NOT POOL INTERVALS.

Cohort	Method	Rank-1 (%)
LFW	FaceNet_sign512	98.486
LFW	SFace_cosine	98.887
LFW	SFace_sign128	96.602
LFW	SFace_RP_centered_512_seed20261030	98.545
LFW	SFace_RP_centered_512_seed20261031	98.525
CelebA	FaceNet_sign512	86.055
CelebA	SFace_cosine	92.285
CelebA	SFace_sign128	83.213
CelebA	SFace_RP_centered_512_seed20261030	89.951
CelebA	SFace_RP_centered_512_seed20261031	90.381
CelebA_external	FaceNet_sign512	85.146
CelebA_external	SFace_cosine	93.691
CelebA_external	SFace_sign128	87.012
CelebA_external	SFace_RP_centered_512_seed20261030	91.846
CelebA_external	SFace_RP_centered_512_seed20261031	91.768

TABLE XVII
SFACE 512 MINUS FACENET 512 RANK-1, PERCENTAGE POINTS. NOMINAL IDENTITY-BOOTSTRAP INTERVALS,128 UNITS AFTER AVERAGING 20 SPLITS; CONDITIONAL ON THE SELECTED IMAGES. BOTH SEEDS AND THE NEAR-ZERO LFW 128 DIFFERENCE ARE SHOWN. DEVELOPMENT/MULTIPLE COMPARISONS LIMIT CONFIRMATORY INTERPRETATION.

Cohort	Seed	Delta (pp)	95% CI (pp)
LFW 128	20261030	0.059	$[-1.026, 1.084]$
LFW 128	20261031	0.039	$[-1.035, 1.006]$
CelebA 128	20261030	3.896	$[1.719, 6.055]$
CelebA 128	20261031	4.326	$[2.383, 6.240]$
CelebA_external 128	20261030	6.699	$[4.609, 8.838]$
CelebA_external 128	20261031	6.621	$[4.570, 8.789]$
LFW 4096	20261030	3.623	$[1.729, 5.449]$
LFW 4096	20261031	3.223	$[1.514, 4.922]$

direction queue: 139994 bytes. Every peer connection uses a task-owned loopback relay; no global network settings are changed. This models stream delay, not packet loss, physical separation or an imposed WAN bandwidth.

TABLE XVIII
INPUT-INCLUSIVE FIVE-PARTY MASKED-SYSTEM REFERENCE: MEAN AND SAMPLE SD OF THREE VM TIMINGS, GLOBAL TRAFFIC (MB) AND LARGEST-CHILD PEAK RSS (MIB). FIVE COLOCATED PROCESSES; NO WAN CLAIM.

N	Operation	Seconds	MB	RSS
8	Re-key	9.82±1.30	471.8	38.7
8	Query	2.28±0.05	10.9	27.6
16	Re-key	10.05±0.35	722.6	48.3
16	Query	1.71±0.25	18.3	32.9
32	Re-key	12.41±1.51	997.3	60.0
32	Query	2.08±0.40	33.2	38.0
64	Re-key	16.77±0.19	1564.0	93.5
64	Query	2.30±0.07	63.4	53.0
128	Re-key	28.54±2.37	2717.4	132.8
128	Query	4.09±0.35	122.5	63.1
256	Re-key	44.02±0.31	5051.5	209.6
256	Query	8.35±0.42	245.0	89.5

TABLE XIX
COMPLETE LIFECYCLE FROM EACH PARTY'S OWN PERSISTED SHARE FILE. MEAN AND SAMPLE SD OF THREE MEASURED VM TIMINGS; GLOBAL MB. NO PLAINTEXT DATABASE/INDEX OR OLD KEYS ARE RESUPPLIED TO A COMPUTING PARTY.

N	State	Operation	Seconds	MB
8	Masked	Re-key	10.73±2.15	470.7
8	Shared root	Re-key	7.84±0.84	453.6
8	Masked	Query	2.24±0.28	10.8
8	Shared root	Query	0.40±0.07	0.9
128	Masked	Re-key	40.76±6.51	2710.0
128	Shared root	Re-key	30.36±4.51	2485.3
128	Masked	Query	6.63±1.55	120.4
128	Shared root	Query	1.22±0.21	14.6

TABLE XX
INPUT-INCLUSIVE SAME-CONFIGURATION BASELINES AND PRIVATE INDEXED-QUERY REFERENCE. THREE MEASURED RUNS AFTER WARMUP; MEAN AND SAMPLE SD. FULL PHASE DETAILS ACCOMPANY THE LOGS.

N	Design	Operation	Seconds	MB
8	Raw roots	Re-key	5.74±0.27	454.8
8	Raw roots	Query	0.50±0.16	1.0
8	Private index	Query	43.28±0.86	4965.0
32	Raw roots	Re-key	11.03±1.15	939.1
32	Raw roots	Query	0.48±0.03	4.2
128	Raw roots	Re-key	22.49±0.54	2492.7
128	Raw roots	Query	0.91±0.05	16.7
8	Compact sort	Re-key	93.88±22.51	7274.6
16	Compact sort	Re-key	127.35±4.56	15788.2

TABLE XXI
MEAN PARTY-ZERO ROUNDS AND LARGEST SINGLE-PARTY PEAK RSS (MIB, MEAN/SAMPLE SD). FIVE MEASURED PERSISTED/CLIENT TRIALS AND THREE COLD-INPUT TRIALS. PEAK MEMORY IS NOT AN AGGREGATE OVER FIVE PARTIES; ALL PER-RUN VM/WALL TIMES AND PHASE TRAFFIC REMAIN IN THE ARTIFACT.

Interface	N	Operation	View	Rounds	Peak RSS
Persisted rekey	128	Rekey	Materialized	28526	137.22 ± 0.16
Persisted rekey	128	Rekey	Virtual	25448	136.92 ± 0.11
Cold	1024	Rekey	Materialized	223717	673.62 ± 0.19
Cold	1024	Rekey	Virtual	202495	677.06 ± 0.13
Cold	1024	Query	Materialized	35070	97.47 ± 0.30
Cold	1024	Query	Virtual	35232	97.11 ± 0.46
Cold	512	Rekey	Materialized	114122	363.53 ± 0.45
Cold	512	Rekey	Virtual	103268	363.37 ± 0.08
Cold	512	Query	Materialized	18234	65.30 ± 0.08
Cold	512	Query	Virtual	18396	65.60 ± 0.37
Private client	128	Query	Full scan	3913	68.33 ± 0.40
Private client	128	Query	Fused	4107	75.87 ± 0.22
Private client	128	Query	Polynomial	18707	93.01 ± 0.09

TABLE XXII
USER-SPACE STREAM-DELAY SENSITIVITY: ONE WARMUP PLUS 3 ALTERNATING MEASURED TRIALS. N2D64 IS REKEY; N8D512 IS QUERY. REQUESTED ADDED RTT, MEAN/SD VM TIME AND GLOBAL MB. SIZES AND OPERATIONS MUST NOT BE POOLED OR EXTRAPOLATED.

N/d	RTT(ms)	View	Seconds	MB
2/64	0	Materialized	7.19 ± 2.87	15.399
2/64	0	Virtual	5.41 ± 1.24	2.180
2/64	1	Materialized	6.72 ± 2.40	15.399
2/64	1	Virtual	4.79 ± 1.36	2.180
2/64	20	Materialized	19.44 ± 0.42	15.399
2/64	20	Virtual	18.67 ± 0.76	2.180
2/64	50	Materialized	41.96 ± 0.23	15.399
2/64	50	Virtual	41.81 ± 0.33	2.180
8/512	0	Materialized	5.99 ± 2.13	10.901
8/512	0	Virtual	7.69 ± 3.46	11.108
8/512	1	Materialized	4.89 ± 0.13	10.901
8/512	1	Virtual	5.44 ± 0.23	11.108
8/512	20	Materialized	25.16 ± 0.41	10.901
8/512	20	Virtual	25.92 ± 0.64	11.108
8/512	50	Materialized	54.04 ± 0.26	10.901
8/512	50	Virtual	58.70 ± 0.62	11.108

References

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